

Minimal Langgraph Mvp

Minimal LangGraph TODO MVP

Goal

Build a small LangGraph workflow that processes typed text or transcribed voice and updates a tenant-scoped TODO list.

Voice or text input -> FastAPI -> optional STT -> LangGraph -> PostgreSQL -> UI response

Minimal Screens

Screen	Fields and controls
Registration/setup	API key, username
Main UI	Text input, microphone button, task list, response area

Tenant Setup

The backend generates `tenant\_id` during registration:

```
tenant_id = "tenant_" + sha256(username)[0:8] + "_" + random_4_digits
```

Every database table includes `tenant\_id`.

Minimal LangGraph Nodes

```
START
|
normalize_input
|
detect_intent
|
extract_entities
|
match_task when needed
|
decide_action
|
execute_tool
|
generate_response
|
END
```

Supported Actions

Intent	Example
create_task	Add gym at 7 PM
update_task	Update meeting time to 4 PM
complete_task	Mark backend task as completed
delete_task	Remove yesterday's pending task
prioritize_task	Make backend task high priority
query_tasks	What did I complete today?

Keep It Simple

- Use one LangGraph workflow with task tools.
- Keep speech-to-text outside the graph.
- Scope every tool call by `tenant\_id`.
- Log every create, update, complete, delete, and query action.
- Ask a clarification question only when task matching is uncertain.